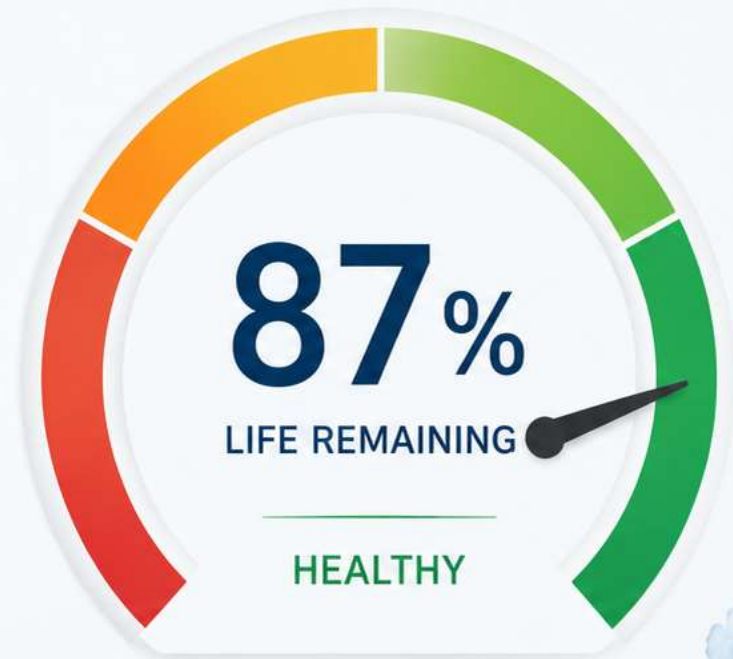


# トランスの じゅみょう メーター Transformer Life Gauge

Real-Time Estimation of Transformer Insulation Life



Turning transformer thermal behavior into an easy-to-understand health indicator.

A presentation by  
Rahul Sahu  
Aditya University

トランスの じょうたい が わかる

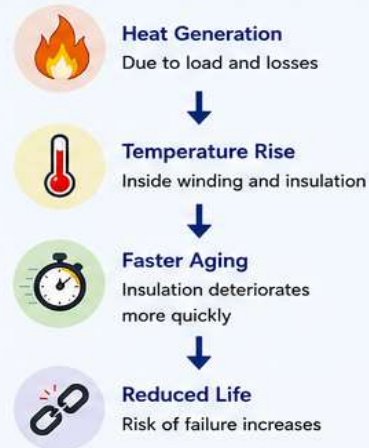
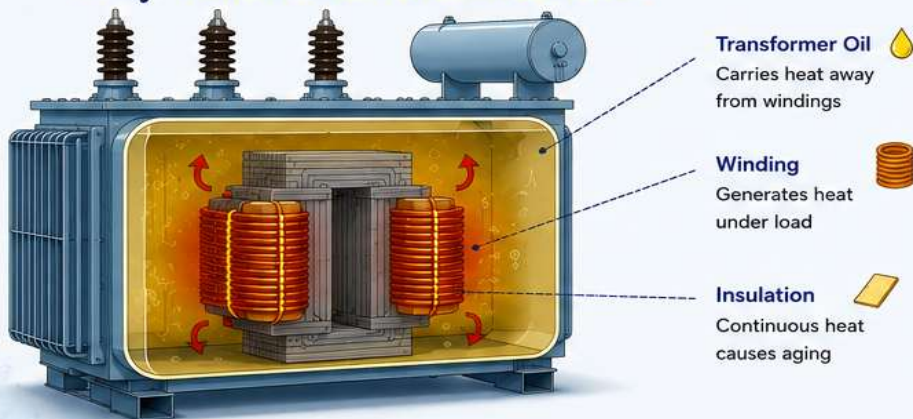
# いまの ほしゅ と じゅみょうの わけ Why Current Maintenance Leads to Life Decrease

Maintenance is time-based, not condition-based. Heat is the main reason for insulation aging.



Maintenance happens at fixed intervals, not based on the real condition of the transformer.

## Why Transformer Life Decreases



## Industry Failure Data

Source: CIGRE Transformer Reliability Survey



25% of known failure causes come from insulation ageing + improper maintenance.

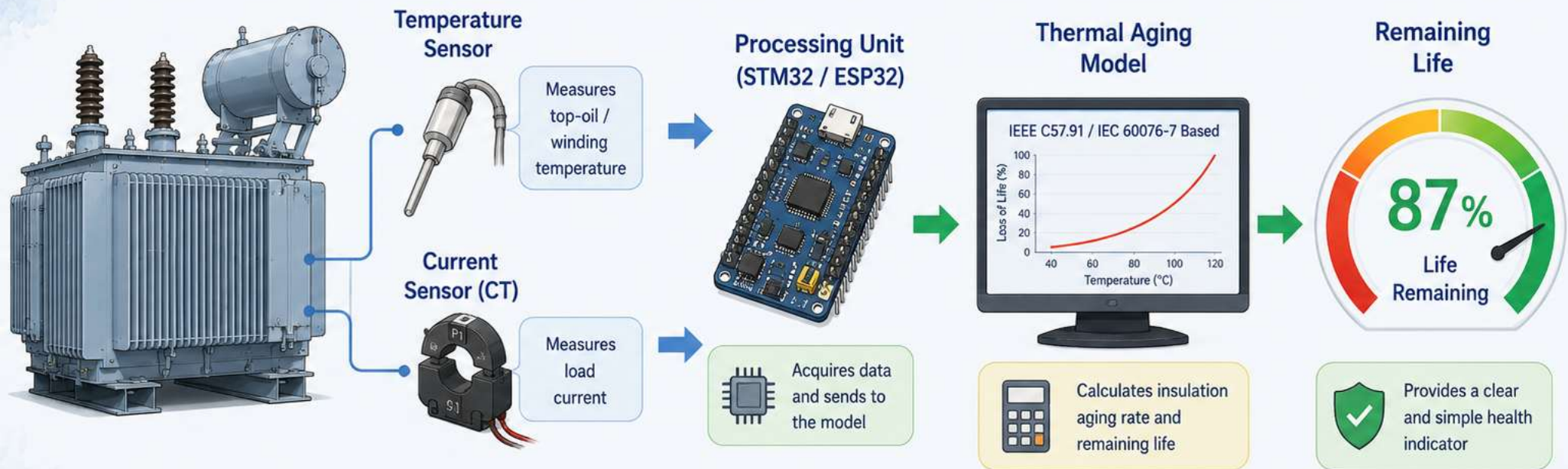
~0.3–0.4% / year  
Overall large power transformer failure rate (CIGRE, modern units).



# あたらしい アイデア

## Transformer Life Gauge Concept

Use key operating parameters and a thermal aging model to estimate remaining insulation life.



- ✓ Combines temperature and load current for accurate life estimation.
- ✓ Based on industry-standard thermal aging principles.
- ✓ Enables real-time assessment of transformer insulation health.

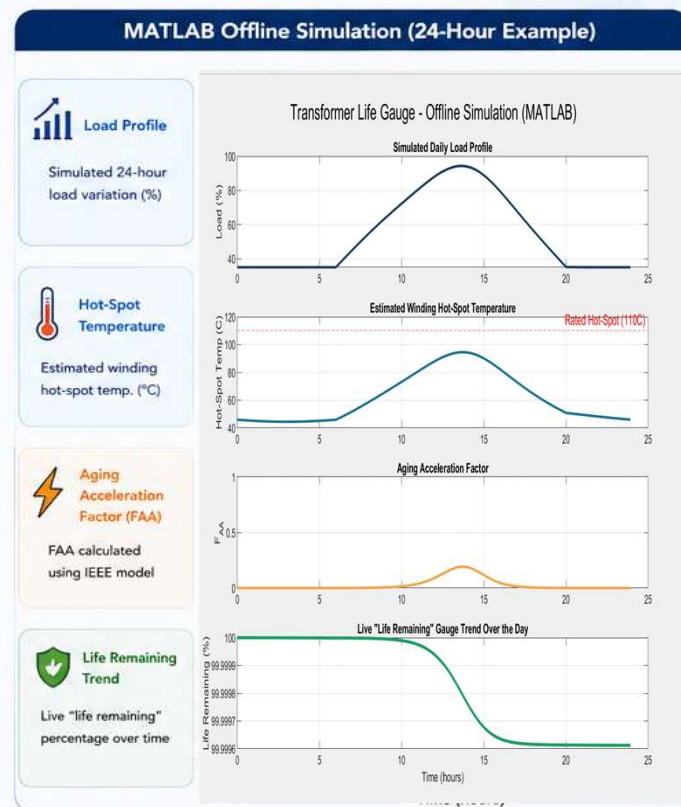




# システムのこうせい Life Calculation Model



Three-step mathematical model to estimate transformer life in real-time.



Physics-Based Model  
(IEEE C57.91)



Accurate Real-Time  
Life Estimation



Smarter Loading  
Decisions



Extends Transformer  
Life & Reliability



## Offline MATLAB Validation

The mathematical model is validated with a simulated daily load profile before deployment on the IoT hardware.



# ライブ モニター Live Monitoring Dashboard

Real-time data visualization and instant alerts for better decision making.

## モニター できるもの What We Monitor



おんど  
Oil / Winding  
Temperature



でんりゅう  
Load Current



あぶらの じょうたい  
Oil Quality



かんきょう  
環境 条件  
Environment



うんてん じかん  
運転時間  
Operating Hours

## Transformer Life Gauge

Online



Overview



Trends



Alerts



Settings

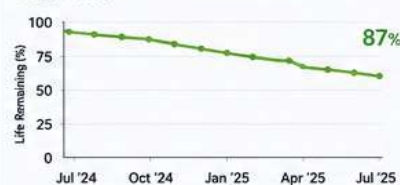
### Life Remaining



### Key Parameters

|                     |         |
|---------------------|---------|
| Oil Temperature     | 62.4 °C |
| Winding Temperature | 68.7 °C |
| Load Current        | 412 A   |
| Oil Quality         | Good    |
| Ambient Temperature | 31.2 °C |

### Life Trend



### Health Status



じょうたい は せいじょう です  
Status is Normal

Last Updated: 12 May 2025, 10:30 AM

どこからでも  
かんたんに チェック!

Easy to check  
from anywhere!



ひとめで わかる  
Easy to Understand  
at a Glance



リアルタイム データ  
Real-Time Data  
Updates



アラートで まえに きづく  
Early Alerts,  
Prevent Failures



データで よりよい  
かんりを サポート  
Data-Driven  
Better Management

# まとめとこれから

## Conclusion & Future Scope

Smarter monitoring today, stronger power tomorrow.

### WHAT WE ACHIEVED



### FUTURE SCOPE



あんぜんな データつうしん  
Secure Data Transmission



リアルタイム モニター  
Real-Time Monitoring



アラートで じこを よぼう  
Early Fault Detection



データで じゅみようを のぼす  
Data-Driven Life Extension





ありがとう  
ございます

Thank You

Your time and attention are  
highly appreciated.



Let's build a smarter  
and safer power future,  
together.



Reliable  
Monitoring



Instant  
Alerts



Smart  
Insights



Better  
Power Future